

Field Theory And Second Quantization

2026.09.01 고유량

QFT study #1

목차

1. Notation and convention
2. Classical field Theory
 1. String recap
 2. Hamiltonian and Lagrangian
 3. Euler – Lagrange Equation
 4. Noether's theorem
3. Quantization of the field
 1. Quantization of the field
 2. Momentum States and Field Operators

Notation and convention

1. Natural units

$$\hbar = c = 1$$

$$[\text{length}] = [\text{time}] = [\text{energy}]^{-1} = [\text{mass}]^{-1}$$

2. metric tensor

$$g_{\mu\nu} = g^{\mu\nu} = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}$$

Notation and convention

3. 4-vector

- position

$$x^\mu = (x^0, \vec{x}), \quad x_\mu = g_{\mu\nu}x^\nu = (x^0, -\vec{x})$$

- momentum

$$p^\mu = (E, \vec{p})$$

- Derivative

$$\partial_\mu = \frac{\partial}{\partial x^\mu} = \left(\frac{\partial}{\partial t}, \vec{\nabla} \right)$$

Notation and convention

4. Lorentz contraction

4-vector $V^\mu = (V^0, \vec{V})$ 에 대해서,

$$V^\mu V_\mu = V_\mu V^\mu = g_{\mu\nu} V^\nu V^\mu = g^{\mu\nu} V_\nu V_\mu = (V^0)^2 - \vec{V}^2 = V^2$$

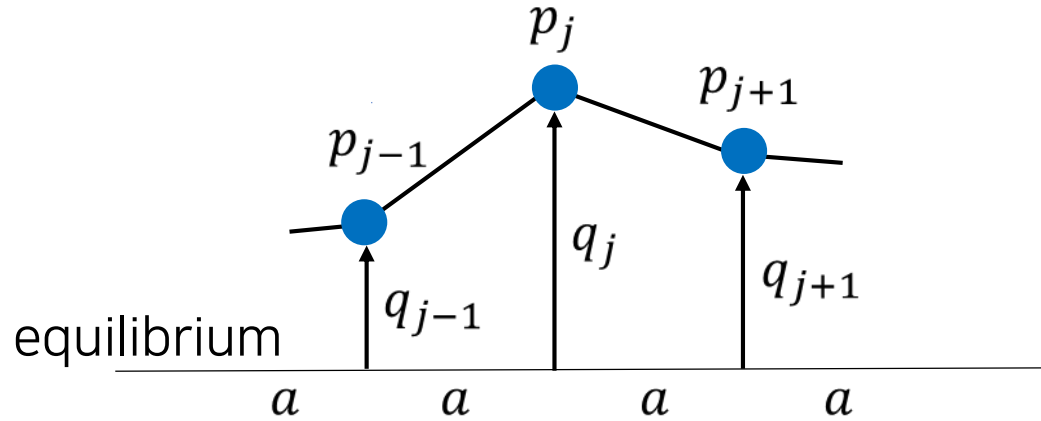
또 다른 4-vector $W^\mu = (W^0, \vec{W})$ 에 대해서

$$V^\mu W_\mu = V_\mu W^\mu = g_{\mu\nu} V^\nu W^\mu = W^0 V^0 - \vec{V} \cdot \vec{W} = VW$$

책에서는 $V^\mu W^\mu$ 혹은 $V_\mu W_\mu$ 과 같은 표기도 쓰지만, 최대한 지양할 예정.

Classical field Theory

Classical field Theory – String recap



$$L = \sum_j \left[\frac{1}{2} M \dot{q}_j^2 - \frac{1}{2} \kappa (q_{j+1} - q_j)^2 \right] \quad p_j = \frac{\partial L}{\partial \dot{q}_j} = M \dot{q}_j$$

$$H = \sum_j p_j \dot{q}_j - L = \sum_j \left[\frac{p_j^2}{2M} + \frac{1}{2} \kappa (q_{j+1} - q_j)^2 \right]$$

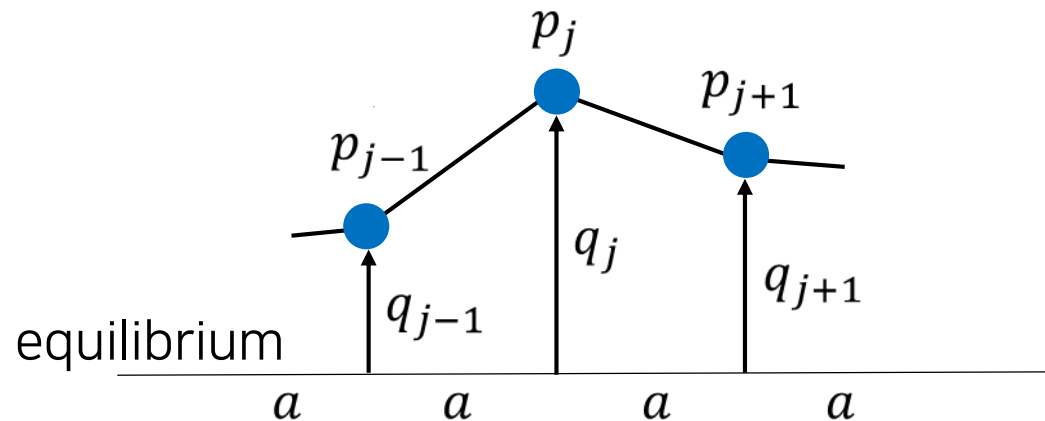
변위 $q_j(t)$ 는 normal mode $\eta_r(t)$ 들의 선형합으로 쓸 수 있다

$$q_j(t) = \sum_r e^{ik_r x_j} \eta_r(t)$$

이때, normal mode는 하나의 SHO처럼 행동한다. 즉,

$$\ddot{\eta}_r + \omega_r^2 \eta_r = 0$$

Classical field Theory – String recap



$$L = \sum_j \left[\frac{1}{2} M \dot{q}_j^2 - \frac{1}{2} \kappa (q_{j+1} - q_j)^2 \right] \quad p_j = \frac{\partial L}{\partial \dot{q}_j} = M \dot{q}_j$$

$$H = \sum_j p_j \dot{q}_j - L = \sum_j \left[\frac{p_j^2}{2M} + \frac{1}{2} \kappa (q_{j+1} - q_j)^2 \right]$$

한편, Continuum limit, $a \rightarrow 0$ 을 하자.

$$q_j(t) \rightarrow q(x, t) \equiv \phi(x, t)$$

$$q_{j+1} - q_j = a \frac{q_{j+1} - q_j}{a} \rightarrow a \partial_x \phi(x, t), \quad \sum_j = \frac{1}{a} \sum_j a \rightarrow \frac{1}{a} \int dx$$

$$L = \int dx \left(\frac{1}{2} \frac{M}{a} (\partial_t \phi)^2 - \frac{1}{2} \kappa a (\partial_x \phi)^2 \right)$$

Classical field Theory – String recap

$$\begin{aligned} L &= \int dx \left(\frac{1}{2} \mu (\partial_t \phi)^2 - \frac{1}{2} T (\partial_x \phi)^2 \right) \\ L &\rightarrow \int d^3x \left(\frac{1}{2} \mu (\partial_t \phi)^2 - \frac{1}{2} T (\vec{\nabla} \phi)^2 \right) \\ &= \mu \int d^3x \left(\frac{1}{2} (\partial_t \phi)^2 - \frac{1}{2} c_s^2 (\vec{\nabla} \phi)^2 \right), \quad \text{where } c_s = \sqrt{\frac{T}{\mu}} \end{aligned}$$

상대론적 이론에서, Lagrangian은 Lorentz scalar 여야 하므로, $c_s = c = 1$ 로 둔다.

$$L = \int d^3x \frac{1}{2} (\partial_\mu \phi)(\partial^\mu \phi)$$

Discrete degree of freedom $q_1, q_2, q_3, \dots$ $\rightarrow$ Continuous set of degree of freedom $\phi(\vec{x}, t)$

Hamiltonian and Lagrangian

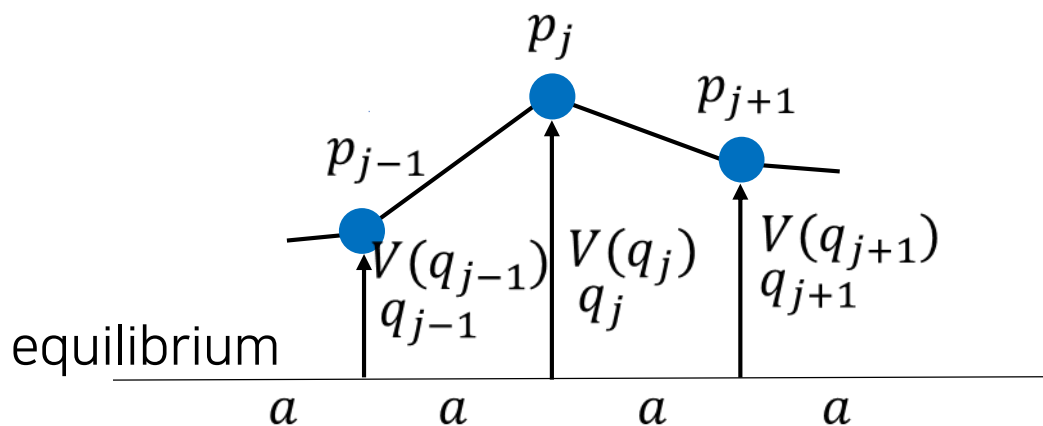
Lagrangian은 Lagrangian density의 공간적분이다.

$$L = \int d^3x \mathcal{L}$$

Canonical conjugate momentum $p_j = \frac{\partial L}{\partial \dot{q}_j} \rightarrow \pi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}}$

$$\begin{aligned} \mathcal{H} &= \pi \dot{\phi} - \mathcal{L} \\ H &= \int d^3x \mathcal{H} \end{aligned}$$

Hamiltonian and Lagrangian



즉,

$$L = \sum_j \left[\frac{1}{2} M \dot{q}_j^2 - \frac{1}{2} \kappa (q_{j+1} - q_j)^2 - V(q_j) \right]$$

$$L \rightarrow \int d^3x \left[\frac{1}{2} (\partial_t \phi)^2 - \frac{1}{2} (\vec{\nabla} \phi)^2 - \mathcal{V}[\phi] \right]$$

$$\mathcal{L} = \frac{1}{2} (\partial_t \phi)^2 - \frac{1}{2} (\vec{\nabla} \phi)^2 - \mathcal{V}[\phi]$$

$$= \frac{1}{2} (\partial_\mu \phi)(\partial^\mu \phi) - \mathcal{V}[\phi]$$

한편, Hamiltonian :

$$\mathcal{H} = \frac{1}{2} \pi^2 + \frac{1}{2} (\vec{\nabla} \phi)^2 + \mathcal{V}[\phi]$$

Hamiltonian and Lagrangian

앞선 예시에서, $\mathcal{K} = \frac{1}{2}(\partial_t \phi)^2 = \frac{1}{2}\pi^2$ 그리고 $\mathcal{V} = \frac{1}{2}(\vec{\nabla}\phi)^2 + \mathcal{V}[\phi]$ 라고 둔다면,

$$\mathcal{L} = \mathcal{K} - \mathcal{V}, \quad \mathcal{H} = \mathcal{K} + \mathcal{V}$$

로 볼 수 있다.

H 는 conserved quantity 이지만($P^\mu = (H, \vec{P})$, $\mathcal{H} = \mathcal{T}^{00}$), Lorentz invariant는 아니다.

L 은 conserved quantity가 아니지만, Lorentz scalar 이므로, relativistic system에서 자주 쓴다.

Hamiltonian and Lagrangian

한편, Field Theory에서는 Kinetic energy, Potential energy라고 쓰지 않는다.
대신, Kinetic terms(bilinear), 그리고 interactions 라고 부른다.

Kinetic terms (bilinear) – Two fields

$$\mathcal{L}_K \supset \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi), \quad \bar{\psi} i \not{\partial} \psi, \quad \frac{1}{4} F_{\mu\nu}^2, \quad \left(\frac{1}{2} m^2 \phi^2 \right), \quad \dots$$

interactions – three or more fields

$$\mathcal{L}_{\text{int}} \supset \lambda \phi^3, \quad e A_\mu \bar{\psi} \gamma^\mu \psi, \quad g^2 A_\mu^2 A_\nu^2, \quad \dots$$

Euler-Lagrange Equation

$$S = \int dt L = \int d^4x \mathcal{L}$$

$\mathcal{L} = \mathcal{L}[\phi, \partial_\mu \phi]$ 라고 두자. 그렇다면,

$$\begin{aligned} \delta S &= \int d^4x \delta \mathcal{L}[\phi, \partial_\mu \phi] \\ &= \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta (\partial_\mu \phi) \right] \\ &= \int d^4x \left\{ \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right] \delta \phi + \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta \phi \right] \right\} \end{aligned}$$

Euler-Lagrange Equation

$$\begin{aligned}\delta S &= \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right] \delta \phi + \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta \phi \right]_{\text{boundary}} \\ &= \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right] \delta \phi\end{aligned}$$

결과적으로, Lagrangian density의 total derivative term은 없앨 수 있다. 즉, 부분적분으로 Lagrangian density 를 다시 적을 수도 있다. 예를 들면,

$$A^\mu \partial_\mu B = -(\partial_\mu A^\mu) B, \quad \frac{1}{2} (\partial_\mu \phi) (\partial^\mu \phi) = -\frac{1}{2} \phi \square \phi$$

Euler-Lagrange Equation

$$\begin{aligned}\delta S &= \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right] \delta\phi + \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta\phi \right]_{\text{boundary}} \\ &= \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right] \delta\phi \\ &= 0\end{aligned}$$

For any field $\delta\phi$ 에서 성립하기 위해선,

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} = 0$$

Euler-Lagrange Equation

예를 들어서

$$\begin{aligned}\mathcal{L} &= \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2}m^2\phi^2 \\ &= \frac{1}{2}g^{\mu\nu}(\partial_\mu \phi)(\partial_\nu \phi) - \frac{1}{2}m^2\phi^2\end{aligned}$$

then

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \phi} &= -m^2\phi \\ \partial_\rho \frac{\partial \mathcal{L}}{\partial(\partial_\rho \phi)} &= \partial_\rho \left[\frac{1}{2}g^{\mu\nu}\delta_\mu^\rho(\partial_\nu \phi) + \frac{1}{2}g^{\mu\nu}(\partial_\mu \phi)\delta_\nu^\rho \right] \\ &= g^{\rho\sigma}\partial_\rho\partial_\sigma\phi = \square\phi\end{aligned}$$

따라서 Equation of motion은

$$(\square + m^2)\phi = 0$$

Noether's Theorem

Noether's Theorem

Lagrangian은 ϕ 를 바꿔도 불변일 수 있다. 예를 들어,

$$\begin{aligned}\mathcal{L} &= |\partial_\mu \phi|^2 - m^2 |\phi|^2 \\ &= (\partial_\mu \phi)(\partial^\mu \phi^*) - m^2 \phi \phi^*\end{aligned}$$

Lagrangian는 변환

$$\phi \rightarrow \phi' = e^{-i\alpha} \phi, \quad \phi^* \rightarrow \phi'^* = e^{i\alpha} \phi$$

에 대해서 대칭이다. (α 는 실수인 상수)

α 는, 복소평면상에서 얼마만큼 회전했는가를 나타내는 parameter이며, 작은 α 를 잡을 수도 있다.

$$e^{-i\alpha} = 1 - i\alpha, \quad e^{i\alpha} = 1 + i\alpha$$

Noether's Theorem

따라서, Lagrangian $\mathcal{L} = \mathcal{L}[\phi_n, \partial_\mu \phi_n]$ 일 때,

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \alpha} &= \sum_n \left[\frac{\partial \mathcal{L}}{\partial \phi_n} \frac{\delta \phi}{\delta \alpha} + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \frac{\delta (\partial_\mu \phi_n)}{\delta \alpha} \right] \\ &= \sum_n \left[\left(\frac{\partial \mathcal{L}}{\partial \phi_n} - \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \right) \frac{\delta \phi_n}{\delta \alpha} + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \frac{\delta \phi_n}{\delta \alpha} \right) \right] \\ &= \partial_\mu \sum_n \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \frac{\delta \phi_n}{\delta \alpha} \right) = 0\end{aligned}$$

Noether current

$$J^\mu = \sum_n \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \frac{\delta \phi_n}{\delta \alpha} \right)$$

Noether's Theorem

예를 들어, $\mathcal{L} = (\partial_\mu \phi)(\partial^\mu \phi^*) - m^2 \phi \phi^*$ 에서는

$$J^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \frac{\delta \phi}{\delta \alpha} + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^*)} \frac{\delta \phi^*}{\delta \alpha} = -i(\phi \partial^\mu \phi^* - \phi^* \partial^\mu \phi)$$

$$\begin{aligned} \partial_\mu J^\mu &= -i[\phi \square \phi^* - \phi^* \square \phi] \\ &= -i[\phi(-m^2 \phi^*) - \phi^*(-m^2 \phi)] \\ &= 0 \end{aligned}$$

Noether's Theorem

특히, $\partial_\mu J^\mu = 0$ 를 만족하는 J^μ 를 conserved current 라고 한다. 다음과 같이 정의된 charge Q

$$Q = \int d^3x J_0$$

에 대해서

Lorenz

$$\partial_t Q = \partial_t \int d^3x J_0 = \int d^3x \vec{\nabla} \cdot \vec{J} = \oint d\vec{a} \cdot \vec{J} = 0$$

Noether's theorem

If a Lagrangian has a continuous symmetry then there exists a current associated with that symmetry that is conserved when the equations of motion are satisfied.

Noether's Theorem

Recall that we needed to assume the symmetry was continuous so that small variations $\frac{dL}{d\alpha}$ could be taken. So, Noether's theorem does not apply to discrete symmetries.

Important points about this theorem are :

- The symmetry must be continuous, otherwise $\delta\alpha$ has no meaning.
- The current is conserved on-shell, that is, when the equations of motion are satisfied.
- It works for global symmetries, parametrized by numbers α , not only for local (gauge) symmetries parametrized by functions $\alpha(x)$.

Energy-momentum Tensor

Lagrangian이 아닌 action에 대한 global symmetry 가 적용되는 Noether's thm의 중요한 경우.

infinitesimal ξ^μ 에 대해 spacetimes transformation

$$\phi(x) \rightarrow \phi(x + \xi) = \phi(x) + \xi^\nu \partial_\nu \phi(x) + \dots$$

즉,

$$\frac{\delta \phi}{\delta \xi^\nu} = \partial_\nu \phi$$

Energy-momentum Tensor

Lagrangian $\mathcal{L}[\phi_n, \partial_\mu \phi_n]$ 에 대해서

$$\begin{aligned}\frac{\delta \mathcal{L}}{\delta \xi^\nu} &= \sum_n \left[\frac{\partial \mathcal{L}}{\partial \phi_n} \frac{\delta \phi_n}{\delta \xi^\nu} + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \frac{\delta (\partial_\mu \phi_n)}{\delta \xi^\nu} \right] \\ &= \sum_n \left[\frac{\partial \mathcal{L}}{\partial \phi_n} \partial_\nu \phi_n + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \partial_\nu \partial_\mu \phi_n \right] \\ &= \partial_\mu \sum_n \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \partial_\nu \phi_n \right] \\ &= \partial_\nu \mathcal{L}\end{aligned}$$

$$\partial_\mu \sum_n \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \partial_\nu \phi_n \right] - \partial_\nu \mathcal{L} = 0$$

Energy-momentum Tensor

$$\partial_\mu \sum_n \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \partial_\nu \phi_n \right] - \partial_\nu \mathcal{L} = 0$$
$$\partial_\mu \sum_n \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \partial_\nu \phi_n - \delta^\mu_\nu \mathcal{L} \right] = 0$$

$\mathcal{T}^\mu_\nu$, energy-momentum tensor

$$\mathcal{T}^\mu_\nu = \sum_n \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \partial_\nu \phi_n - \delta^\mu_\nu \mathcal{L} \right]$$

에 대해

$$\partial_\mu \mathcal{T}^\mu_\nu = 0$$

Energy-momentum Tensor

중요한 성분인 energy density :

$$\mathcal{E} = \mathcal{T}^{00} = \sum_n \left[\frac{\partial \mathcal{L}}{\partial \dot{\phi}} \dot{\phi} - \mathcal{L} \right]$$

또, 대응되는 conserved charge $\mathcal{P}^\mu$ 는

$$\mathcal{P}^\nu = \int d^3x \mathcal{T}^{0\nu} = (H, \vec{P})$$

Current

1. Noether current

$$\partial_\mu J^\mu = 0$$

2. External current

$$J^\mu(x) = (\rho(x), \vec{v}(x))$$

3. Source for a field

$$\mathcal{L} = \dots - A^\mu J_\mu$$

4. Place-holder

$$\begin{aligned}\mathcal{L} &= -\frac{1}{4}F_{\mu\nu}^2 - \phi^* \square \phi - ieA^\mu (\phi^* \partial_\mu \phi - \phi \partial_\mu \phi^*) \\ &= -\frac{1}{4}F_{\mu\nu}^2 - \phi^* \square \phi - A^\mu J_\mu\end{aligned}$$

Green's function (propagator)

Coulomb potential

The best way to understand classical field theory is by doing some calculations.

external current :

$$J^\mu(x) = (e\delta^3(\vec{x}), 0),$$

The Lagrangian is

$$\begin{aligned}\mathcal{L} &= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - A^\mu J_\mu \\ &= -\frac{1}{4}(\partial_\mu A_\nu - \partial_\nu A_\mu)(\partial^\mu A^\nu - \partial^\nu A^\mu) - A^\mu J_\mu \\ &= \frac{1}{2}A^\mu(g_{\mu\nu}\square - \partial_\mu\partial_\nu)A^\nu - A^\mu J_\mu\end{aligned}$$

EoM is

$$\square A_\mu - \partial_\mu\partial_\nu A^\nu = J_\mu$$

Coulomb potential

Lorenz gauge : $\partial_\nu A^\nu = 0$

$$\begin{aligned}\square A_\mu &= J_\mu \\ \rightarrow A_\mu &= \frac{1}{\square} J_\mu\end{aligned}$$

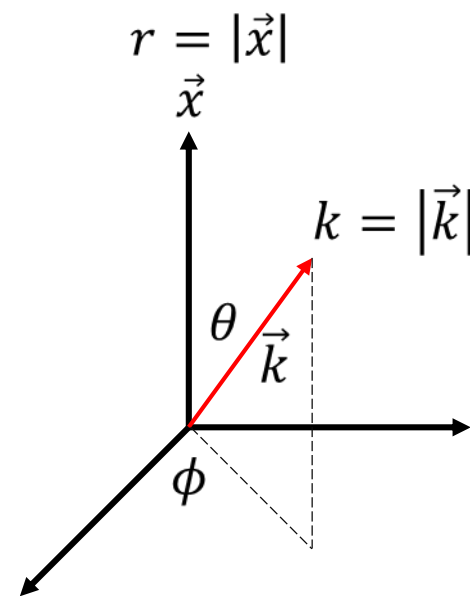
$\frac{1}{\square}$ 는 $\square$ 의 역연산자(inverse operator)

Propagator $\Pi = \frac{1}{\square}$ 로 propagation 하는 source J_μ 가 A_μ 를 결정한다.

Coulomb potential

$\vec{j} = 0$ 이므로 0th 성분만 보면,

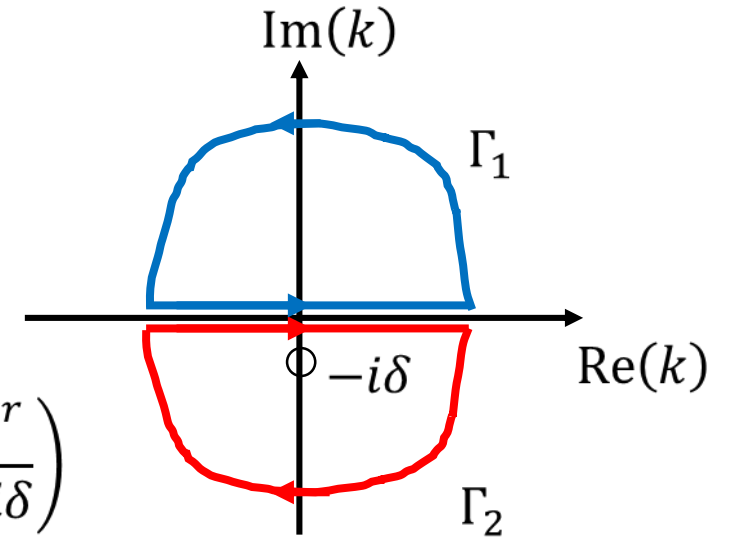
$$\begin{aligned} A_0(x) &= \frac{e}{\square} \delta^3(\vec{x}) \\ &= -\frac{e}{\Delta} \int \frac{d^3k}{(2\pi)^3} e^{i\vec{k}\vec{x}} \\ &= \frac{e}{(2\pi)^3} \int_0^\infty k^2 dk \int_{-1}^1 d\cos\theta \int_0^{2\pi} d\phi \frac{1}{k^2} e^{ikr \cos\theta} \\ &= \frac{e}{(2\pi)^2} \int_0^\infty dk \frac{e^{ikr} - e^{-ikr}}{ikr} \\ &= \frac{e}{8\pi^2} \frac{1}{ir} \int_{-\infty}^\infty dk \frac{e^{ikr} - e^{-ikr}}{k} \end{aligned}$$



Coulomb potential

$$\begin{aligned}
 A_0(x) &= \frac{e}{8\pi^2} \frac{1}{ir} \int_{-\infty}^{\infty} dk \frac{e^{ikr} - e^{-ikr}}{k} \\
 &= \frac{e}{8\pi^2} \frac{1}{ir} \int_{\Gamma} dk \frac{e^{ikr} - e^{-ikr}}{k + i\delta} \\
 &= \frac{e}{8\pi^2} \frac{1}{ir} \left(\int_{\Gamma_1} dk \frac{e^{ikr}}{k + i\delta} - \int_{\Gamma_2} dk \frac{e^{-ikr}}{k + i\delta} \right) \\
 &= \frac{e}{8\pi^2} \frac{2\pi i}{ir} (+e^{-\delta r})
 \end{aligned}$$

$$(\because) A_0(x) = \frac{e}{4\pi r}$$



Green's function

Lagrangian

$$\begin{aligned}\mathcal{L} &= \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \frac{1}{2}m^2\phi^2 + J\phi \\ &= -\frac{1}{2}\phi(\square + m^2)\phi + J\phi\end{aligned}$$

EoM is

$$(\square + m^2)\phi(x) = J(x) \rightarrow \phi(x) = \phi_0(x) + \frac{1}{\square + m^2}J(x)$$

where

$$(\square + m^2)\phi_0(x) = 0$$

Green's function

Define Green's function $\Pi(x, y)$:

$$(\square_x + m^2)\Pi(x, y) = -\delta^4(x - y)$$

Then we can find $\phi(x)$

$$\phi(x) = \phi_0(x) - \int d^4y \Pi(x, y)J(y)$$



Since

$$\left(\frac{1}{\square + m^2}J\right)(x) = - \int d^4y \Pi(x, y)J(y)$$

Green's function

And we have

$$(\square_x + m^2)\Pi(x, y) = -\delta^4(x - y)$$
$$\Pi(x, y) = \int \frac{d^4p}{(2\pi)^4} \frac{1}{p^2 - m^2} e^{ip(x-y)}$$

만약 Green's function 의 정의를

$$(\square_x + m^2)\Pi(x, y) = -i\delta^4(x - y)$$

로 두고 boundary condition을 적절하게 고르면

$$D_F(x, y) = \int \frac{d^4p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\epsilon} e^{ip(x-y)}$$

Green's function

가령, ϕ^3 과 같은 interaction term 이 있다고 해보자.

$$\mathcal{L} = -\frac{1}{2}\phi(\square + m^2)\phi + \frac{1}{3}\lambda\phi^3 + J\phi$$

EoM

$$(\square + m^2)\phi = \lambda\phi^2 + J$$

λ 에 perturbatively 풀자.

$$\phi = \phi^{(0)} + \lambda\phi^{(1)} + \lambda^2\phi^{(2)} + \dots$$

Green's function

$$(\square + m^2) \left(\sum_{n=0} \lambda^n \phi^{(n)} \right) = \sum_{\substack{n=0 \\ m=0}} \lambda^{n+m+1} \phi^{(n)} \phi^{(m)} + J$$

λ^0 :

$$(\square + m^2) \phi^{(0)} = J \rightarrow \phi^{(0)} = \frac{1}{\square + m^2} J$$

Homogeneous solution $\phi_0 = 0$ 로 두었다.

λ^1 , $(n, m) = (0, 0)$:

$$(\square + m^2) \phi^{(1)} = (\phi^{(0)})^2 \rightarrow \phi^{(1)} = \frac{1}{\square + m^2} \left(\frac{1}{\square + m^2} J \right) \left(\frac{1}{\square + m^2} J \right)$$

Green's function

따라서

$$\begin{aligned}\phi(x) &= \frac{1}{\square + m^2}J + \lambda \frac{1}{\square + m^2} \left(\frac{1}{\square + m^2}J \right) \left(\frac{1}{\square + m^2}J \right) + \mathcal{O}(\lambda^3) \\ &= - \int d^4y \Pi(x, y)J(y) \\ &\quad - \lambda \int d^4w d^4y d^4z \Pi(x, w) \Pi(w, y) \Pi(w, z) J(y) J(z)\end{aligned}$$

$$= \begin{array}{c} x \\ \bullet \end{array} \text{--- wavy ---} \star J(y) + \begin{array}{c} x \\ \bullet \end{array} \text{--- wavy ---} \begin{array}{c} w \\ \bullet \end{array} \left\{ \begin{array}{l} \nearrow \star J(z) \\ \searrow \star J(y) \end{array} \right. + \dots$$

Quantization of the field

Second quantization

주어진 Lagrangian

$$\mathcal{L} = \frac{1}{2} (\partial_\mu \phi)^2 - \frac{1}{2} m^2 \phi^2$$

에 대해서, EOM은 다음과 같다.

$$(\square + m^2)\phi(x) = 0$$

Homogeneous solution $\phi_0(x)$ 이지만, 편의를 위해 잠시 $\phi(x)$ 로 적는다.
general solution은 normal mode의 선형합

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3} \eta_p(t) e^{i\vec{p}\vec{x}}$$

변위 $q_j(t)$ 는 normal mode $\eta_r(t)$ 들의 선형합으로 쓸 수 있다

$$q_j(t) = \sum_r e^{ik_r x_j} \eta_r(t)$$

Second quantization

normal mode $\eta_p(t)$ 는 하나의 SHO이다.

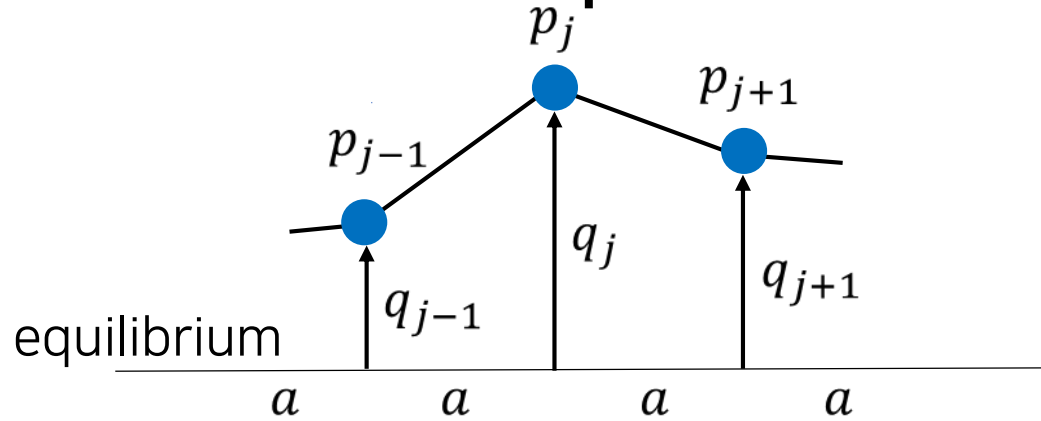
$$\ddot{\eta}_p + \omega_p^2 \eta_p = 0, \quad \text{where } \omega_p^2 = \vec{p}^2 + m^2$$

즉, $\eta_p(t) \sim a_p e^{i\omega_p t} + a_p^* e^{-i\omega_p t}$, ($\phi^* = \phi$ 라는 real scalar 조건으로 a_p, a_p^* 가 됨.)

따라서

$$\begin{aligned} \phi(x) &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (a_p e^{-ipx} + a_p^* e^{ipx}) \\ \pi(x) &= \partial_t \phi(x) = -i \int \frac{d^3p}{(2\pi)^3} \sqrt{\frac{\omega_p}{2}} (a_p e^{-ipx} - a_p^* e^{ipx}) \end{aligned}$$

Second quantization



$$[q_i(t_0), p_j(t_0)] = i\delta_{ij} \rightarrow [\phi(\vec{x}, t_0), \pi(\vec{y}, t_0)] = i\delta^3(\vec{x} - \vec{y})$$

promote $\phi(x), \pi(x)$ as operator

At reference time t_0 , $e^{\pm i\omega_p t}$ are absorbed to a_p , $a_p^\dagger$

$$\phi(\vec{x}) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (a_p e^{i\vec{p}\vec{x}} + a_p^\dagger e^{-i\vec{p}\vec{x}})$$

$$\pi(\vec{x}) = -i \int \frac{d^3p}{(2\pi)^3} \sqrt{\frac{\omega_p}{2}} (a_p e^{i\vec{p}\vec{x}} - a_p^\dagger e^{-i\vec{p}\vec{x}})$$

Second quantization

$$\begin{aligned} [\phi(\vec{x}), \pi(\vec{y})] &= i\delta^3(\vec{x} - \vec{y}) \\ &= \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3k}{(2\pi)^3} \left(-\frac{i}{2}\right) \sqrt{\frac{\omega_p}{\omega_k}} [a_p e^{i\vec{p}\vec{x}} + a_p^\dagger e^{-i\vec{p}\vec{x}}, a_k e^{i\vec{k}\vec{y}} - a_k^\dagger e^{-i\vec{k}\vec{y}}] \\ &= \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3k}{(2\pi)^3} \frac{i}{2} \sqrt{\frac{\omega_p}{\omega_k}} \left(e^{i\vec{p}\vec{x} - i\vec{k}\vec{y}} [a_p, a_k^\dagger] + e^{-i\vec{p}\vec{x} + i\vec{k}\vec{y}} [a_k, a_p^\dagger] \right) \\ &= i \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3k}{(2\pi)^3} \sqrt{\frac{\omega_p}{\omega_k}} e^{i\vec{p}\vec{x} - i\vec{k}\vec{y}} [a_p, a_k^\dagger] \\ &= i \int \frac{d^3p}{(2\pi)^3} e^{i\vec{p}(\vec{x} - \vec{y})} \end{aligned}$$

so, canonical relation give us $[a_p, a_k^\dagger] = (2\pi)^3 \delta^3(\vec{p} - \vec{k})$.

similarly, $[\phi(\vec{x}), \phi(\vec{y})] = 0$

Second quantization

Meanwhile the Hamiltonian density is,

$$\mathcal{H} = \pi \dot{\phi} - \mathcal{L} = \frac{1}{2} \pi^2 + \frac{1}{2} (\nabla \phi)^2 + \frac{1}{2} m^2 \phi^2$$

so,

$$\begin{aligned} H &= \int d^3x \mathcal{H} = \int d^3x \left(\frac{1}{2} \pi^2 + \frac{1}{2} (\nabla \phi)^2 + \frac{1}{2} m^2 \phi^2 \right) \\ &= \int d^3x \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3k}{(2\pi)^3} \left(-\frac{\sqrt{\omega_p \omega_k}}{4} (a_p e^{i\vec{p}\vec{x}} - a_p^\dagger e^{-i\vec{p}\vec{x}}) (a_k e^{i\vec{k}\vec{x}} - a_k^\dagger e^{-i\vec{k}\vec{x}}) \right. \\ &\quad \left. - \frac{\vec{p}\vec{k}}{4\sqrt{\omega_p \omega_k}} (a_p e^{i\vec{p}\vec{x}} - a_p^\dagger e^{-i\vec{p}\vec{x}}) (a_k e^{i\vec{k}\vec{x}} - a_k^\dagger e^{-i\vec{k}\vec{x}}) \right. \\ &\quad \left. + \frac{m^2}{4\sqrt{\omega_p \omega_k}} (a_p e^{i\vec{p}\vec{x}} + a_p^\dagger e^{-i\vec{p}\vec{x}}) (a_k e^{i\vec{k}\vec{x}} + a_k^\dagger e^{-i\vec{k}\vec{x}}) \right) \end{aligned}$$

Second quantization

aa term :

$$\int d^3x \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3k}{(2\pi)^3} \left(-\frac{\sqrt{\omega_p \omega_k}}{4} - \frac{\vec{p}\vec{k}}{4\sqrt{\omega_p \omega_k}} + \frac{m^2}{4\sqrt{\omega_p \omega_k}} \right) e^{i(\vec{p}+\vec{k})\vec{x}} a_p a_k = 0$$

similarly, $a^\dagger a^\dagger$ term :

$$\int d^3x \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3k}{(2\pi)^3} \left(-\frac{\sqrt{\omega_p \omega_k}}{4} - \frac{\vec{p}\vec{k}}{4\sqrt{\omega_p \omega_k}} + \frac{m^2}{4\sqrt{\omega_p \omega_k}} \right) e^{-i(\vec{p}+\vec{k})\vec{x}} a_p^\dagger a_k^\dagger = 0$$

Second quantization

$aa^\dagger$ term :

$$\int d^3x \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3k}{(2\pi)^3} \left(\frac{\sqrt{\omega_p \omega_k}}{4} + \frac{\vec{p} \vec{k}}{4\sqrt{\omega_p \omega_k}} + \frac{m^2}{4\sqrt{\omega_p \omega_k}} \right) e^{i(\vec{p} - \vec{k}) \vec{x}} a_p a_k^\dagger = \int \frac{d^3p}{(2\pi)^3} \frac{\omega_p}{2} a_p a_p^\dagger$$

$a^\dagger a$ term :

$$\int d^3x \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3k}{(2\pi)^3} \left(\frac{\sqrt{\omega_p \omega_k}}{4} + \frac{\vec{p} \vec{k}}{4\sqrt{\omega_p \omega_k}} + \frac{m^2}{4\sqrt{\omega_p \omega_k}} \right) e^{i(\vec{k} - \vec{p}) \vec{x}} a_p^\dagger a_k = \int \frac{d^3p}{(2\pi)^3} \frac{\omega_p}{2} a_p^\dagger a_p$$

Second quantization

with Hamiltonian

$$H_0 = \int \frac{d^3p}{(2\pi)^3} \frac{\omega_p}{2} (a_p^\dagger a_p + a_p a_p^\dagger) = \int \frac{d^3p}{(2\pi)^3} \omega_p \left(a_p^\dagger a_p + \frac{1}{2} [a_p, a_p^\dagger] \right) = \int \frac{d^3p}{(2\pi)^3} (\text{SHO of mode } \vec{p})$$

it is easy to evaluate the commutators

$$[H_0, a_p^\dagger] = \omega_p a_p^\dagger, \quad [H_0, a_p] = -\omega_p a_p$$

where $\omega_p = \sqrt{\vec{p}^2 + m^2}$

Quantum field theory is just quantum mechanics with an infinite number of harmonic oscillators.

Second quantization

비슷하게

$$\vec{P} = - \int d^3x \pi \vec{\nabla} \phi = \int \frac{d^3p}{(2\pi)^3} \vec{p} a_p^\dagger a_p$$

따라서

$$[\vec{P}, a_p^\dagger] = \vec{p} a_p^\dagger, \quad [\vec{P}, a_p] = -\vec{p} a_p$$

Second quantization

Eigenvalue가 E_0 인 Free Hamiltonian의 vacuum state $|0\rangle$ 에서

$$H_0|0\rangle = E_0|0\rangle$$

$$H_0 a_p^\dagger |0\rangle = [H_0, a_p^\dagger] |0\rangle + a_p^\dagger H_0 |0\rangle = (E_0 + \omega_p) a_p^\dagger |0\rangle$$

그리고 통상적으로 vacuum이 translation에 charge를 가지지 않을 때, 즉 $e^{-i\vec{P}\vec{x}}|0\rangle = |0\rangle$ 따라서

$$\vec{P}|0\rangle = 0$$

$$\vec{P} a_p^\dagger |0\rangle = [\vec{P}, a_p^\dagger] |0\rangle + a_p^\dagger \vec{P} |0\rangle = \vec{p} a_p^\dagger |0\rangle$$

Second quantization

따라서 $a_p^\dagger$ 는 vacuum에 energy $\omega_p = \sqrt{\vec{p}^2 + m^2}$, momentum $\vec{p}$ 를 갖는 excitation을 갖게 한다.

$$a_p^\dagger |0\rangle = \frac{1}{\sqrt{2\omega_p}} |\vec{p}\rangle$$

$$H_0 |\vec{p}\rangle = (E_0 + \omega_p) |\vec{p}\rangle, \quad \vec{P} |\vec{p}\rangle = \vec{p} |\vec{p}\rangle$$

이 excitation을 mass m 을 가지는 one particle state라고 해석한다.

Second quantization

There are two new features in second quantization:

1. We have many quantum mechanical systems – one for each $\vec{p}$ – all at the same time.
2. We interpret the n th excitation of the $\vec{p}$ harmonic oscillator as having n **particles**.

Let us take a moment to appreciate this second point.

$$a_{p_1}^\dagger a_{p_2}^\dagger \cdots a_{p_n}^\dagger |0\rangle = \frac{1}{\sqrt{2\omega_{p_1} 2\omega_{p_2} \cdots 2\omega_{p_n}}} |\vec{p}_1, \vec{p}_2, \cdots, \vec{p}_n\rangle$$

Second quantization

Second quantization에서, Hilbert space가 Fock space 로 승격된다.

$$\mathcal{F} = \mathcal{H}_0 \oplus \mathcal{H}_1 \oplus \mathcal{H}_2 \oplus \dots$$

where

$$\begin{aligned}\mathcal{H}_0 &= \text{span}\{|0\rangle\}, \\ \mathcal{H}_1 &= \text{span}\{a_{p_1}^\dagger |0\rangle\}, \\ \mathcal{H}_2 &= \text{span}\{a_{p_1}^\dagger a_{p_2}^\dagger |0\rangle\}, \dots\end{aligned}$$

한편, Hilbert space의 momentum state $|p\rangle$ 는 임의의 4-vector가 아닌, on-shell 4-momentum이다. 따라서 $|p\rangle, |\vec{p}\rangle, |p^\mu\rangle$ 는 같은 뜻으로 쓰인다.

또한, 시간이 지나도, 계를 바꾸어도 Fock space는 불변이다.

Momentum States and Field Operators

one particle state

$$a_p^\dagger |0\rangle = \frac{1}{\sqrt{2\omega_p}} |p\rangle$$

with $[a_p, a_k^\dagger] = (2\pi)^3 \delta^3(\vec{p} - \vec{k})$,

$$\begin{aligned}\langle p|k\rangle &= \sqrt{2\omega_p}\sqrt{2\omega_k}\langle 0|a_p a_k^\dagger|0\rangle = \sqrt{2\omega_p}\sqrt{2\omega_k}[\langle 0|[a_p, a_k^\dagger]|0\rangle + \langle 0|a_k^\dagger a_p|0\rangle] \\ &= 2\omega_p(2\pi)^3 \delta^3(\vec{p} - \vec{k})\end{aligned}$$

당연하게도 $|p\rangle$ 는 orthogonal set이다.

Momentum States and Field Operators

Completeness property for one particle states

$$1 = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} |p\rangle\langle p|$$

check

$$1|k\rangle = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} |p\rangle\langle p|k\rangle = \int d^3p \delta^3(\vec{p} - \vec{k}) |p\rangle = |k\rangle$$

Momentum States and Field Operators

Field operator of free theory

$$\phi_0(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (a_p e^{-ipx} + a_p^\dagger e^{ipx})$$

$$\begin{aligned}\phi_0(x)|0\rangle &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} a_p^\dagger e^{ipx} |0\rangle \\ &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} e^{ipx} |p\rangle\end{aligned}$$

therefore

$$\langle p | \phi_0(x) | 0 \rangle = e^{ipx}$$

Momentum States and Field Operators

QM에서의

$$\langle p|x\rangle = e^{ipx}$$

와 비교하면, $\phi_0(x)|0\rangle = |x\rangle$. 즉, $\phi_0(x)$ 는 position x 에 particle을 하나 만든다. one-point particle $|p\rangle$ 에 대해서 Fock space 안에 $\langle p|\psi\rangle = e^{ipx}$ 를 만족하는 $|\psi\rangle$ 는 무수히 많다.

그러나 one-particle sector 안에서만 $\langle p|\psi\rangle = e^{ipx}$ 를 만족하는 $|\psi\rangle$ 는 $\phi_0(x)|0\rangle$ 가 유일하므로 $\phi_0(x)|0\rangle = |x\rangle$ 라고 해석한다.

Momentum States and Field Operators

한편, 우리 free scalar field $\phi_0(x)$ 는 free Hamiltonian(no interactions) 로 time evolution한다.

$$\begin{aligned}[H, \phi_0(x)] &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} ([H_0, a_p] e^{ipx} + [H_0, a_p^\dagger] e^{-ipx}) \\ &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (-\omega_p a_p e^{ipx} + \omega_p a_p^\dagger e^{-ipx}) \\ &= -i\partial_t \phi_0(x)\end{aligned}$$

Interactions 이 있어도, full Hamiltonian H 에 대해서

$$[H, \phi] = -i\partial_t \phi$$

가 성립한다.

Momentum States and Field Operators

이걸 exact 하게 푸는 경우는 거의 없다. 대신, free field의 notation을 빌려와 쓰게 된다.

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (a_p(t)e^{ipx} + a_p^\dagger(t)e^{-ipx})$$

어떤 한 시각 t_0 에서도 $[a_p(t_0), a_k^\dagger(t_0)] = (2\pi)^3 \delta^3(\vec{p} - \vec{k})$ 을 만족한다. Fock space는 시간이 흘러도 불변이다.

따라서 특정 시각 t_0 을 골라 $a_p(t_0) = a_p$ 라고 정의할 수 있다.

그 시각에서는 $\phi_0(x) = \phi(x)$ 라고 둘 수 있다. 하지만, a_p 의 시간발전은 복잡하며, exact하게 풀 수 없다. 대신, $H = H_0 + H_{\text{int}}$ 로 두고, QM의 time dependent perturbation theory처럼 푼다.